Zwitterionic Surface Charge Mixing as a Predictor of Protein Thermal Stability

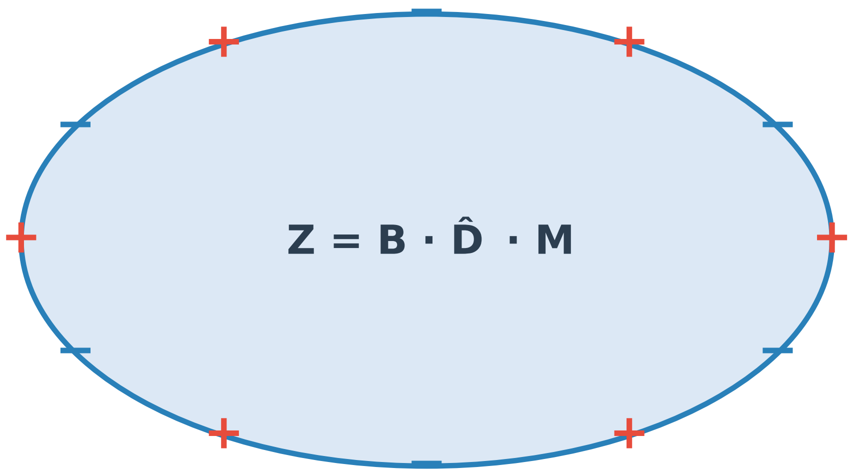
A Cross-Species Analysis

13 organisms · OGT 15–70°C · 34,231 AlphaFold2 protein structures

Meltome Atlas (Jarzab et al., Nature Methods 2020) | ProSurf Z Score

June 2026

The ProSurf Zwitterionic Surface Score



Balance × Density × Mixing

B Balance: $\min(n_+, n_-) / \max(n_+, n_-)$

D̂ Density: charged residues / SASA [normalised]

M Mixing: fraction of opposite-sign neighbours

Research Questions

H1 Do organisms adapted to higher OGT have proteomes with higher mean Z?

H2 Within each organism, do thermally stable proteins have higher Z?

H3 Is any Z–Tm relationship independent of bulk charge composition?

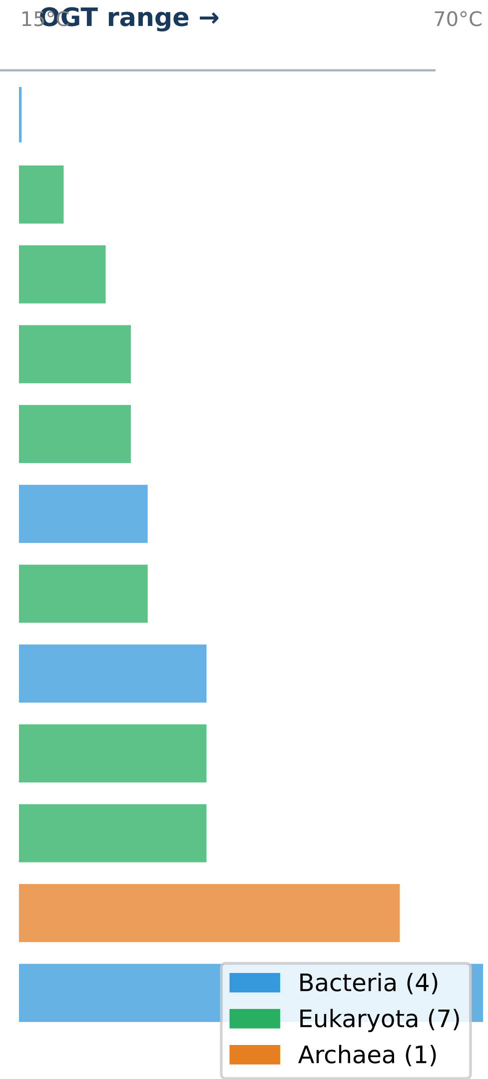
Data sources

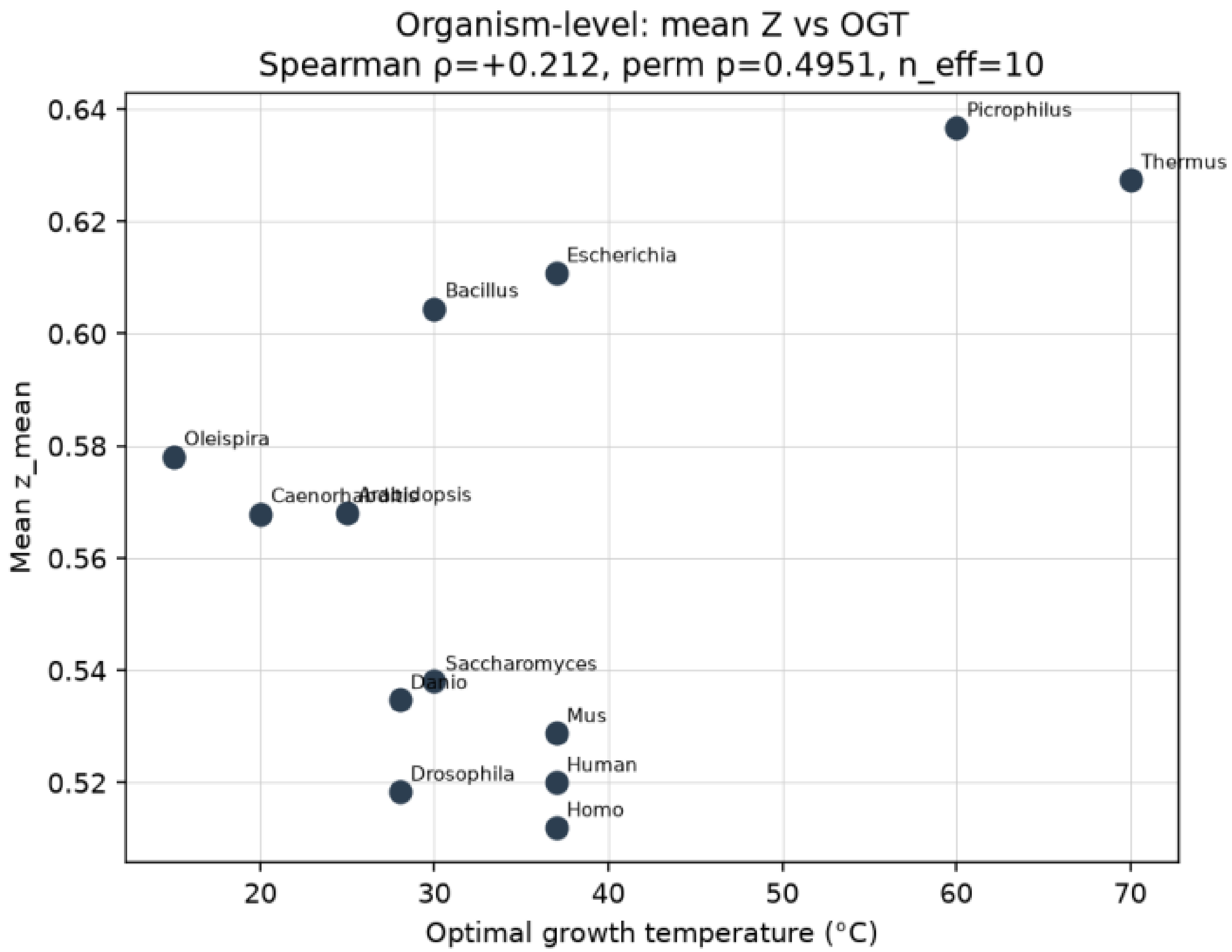
Meltome Atlas — 13 organisms, per-protein Tm (TPP)

AlphaFold2 v6 structures via EBI REST API

34,231 proteins after quality filtering (≥150 residues)

Organism (italic)	OGT (°C)	# Proteins	Domain	OGT range →
• *Oleispira antarctica*	**15**	1,261	Bacteria	
• *Caenorhabditis elegans*	**20**	3,032	Eukaryota	
• *Arabidopsis thaliana*	**25**	2,380	Eukaryota	
• *Danio rerio*	**28**	1,786	Eukaryota	
• *Drosophila melanogaster*	**28**	1,560	Eukaryota	
• *Bacillus subtilis*	**30**	1,354	Bacteria	
• *Saccharomyces cerevisiae*	**30**	1,757	Eukaryota	
• *Escherichia coli*	**37**	1,987	Bacteria	
• *Homo sapiens*	**37**	10,695	Eukaryota	
• *Mus musculus*	**37**	6,434	Eukaryota	
• *Picrophilus torridus*	**60**	767	Archaea	
• *Thermus thermophilus*	**70**	1,118	Bacteria	

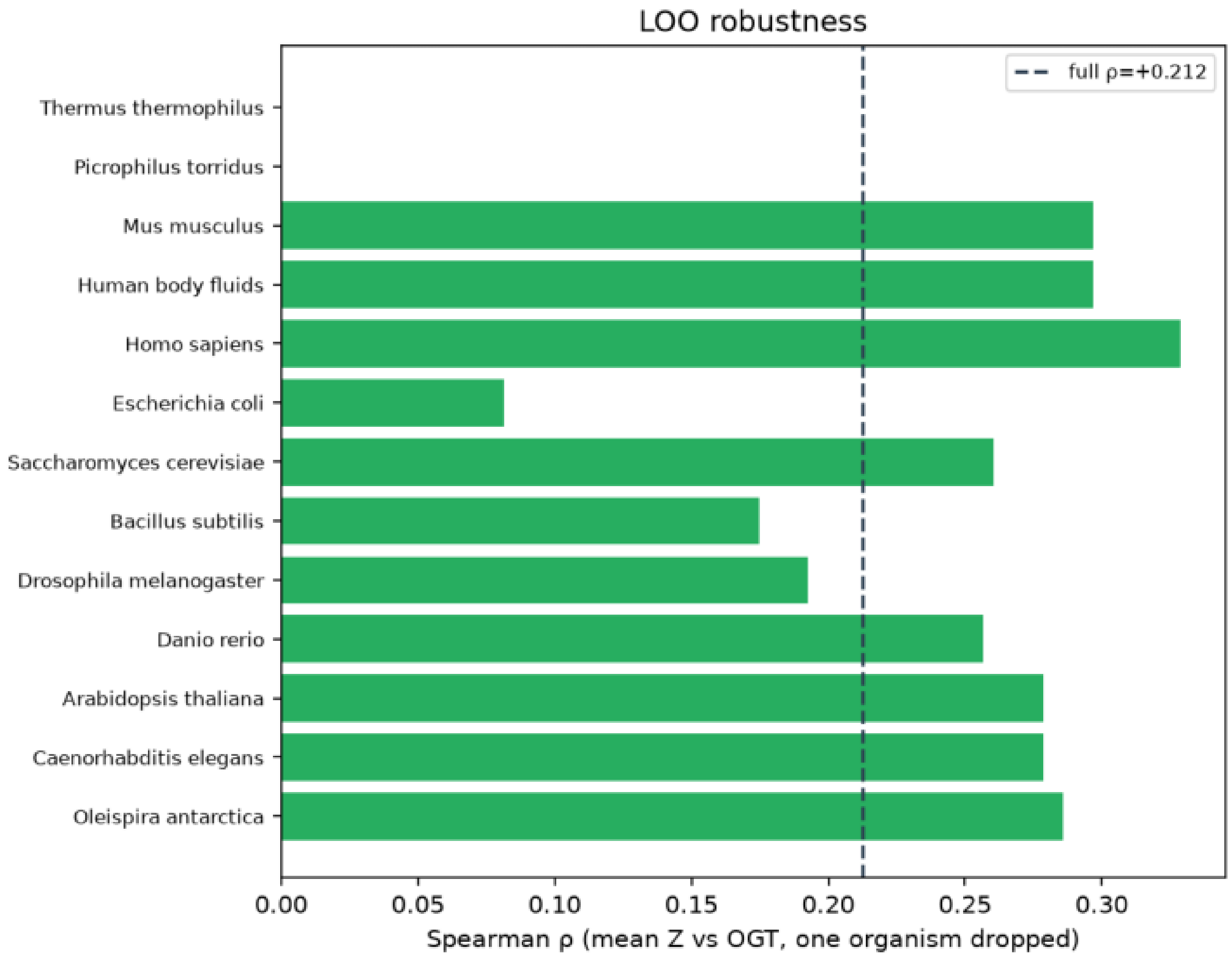




Spearman ρ(mean Z, OGT) = +0.21, perm-p = 0.50

- N = 13 organisms, OGT 15–70°C
- Permutation test with 10,000 shuffles
- Consistent across z_mean / z_max / z_frac (ρ = +0.15–0.21)
- Stable across MIN_PER_ORG cutoffs 30–200
- 10 effective phylogenetic groups — inherently low power

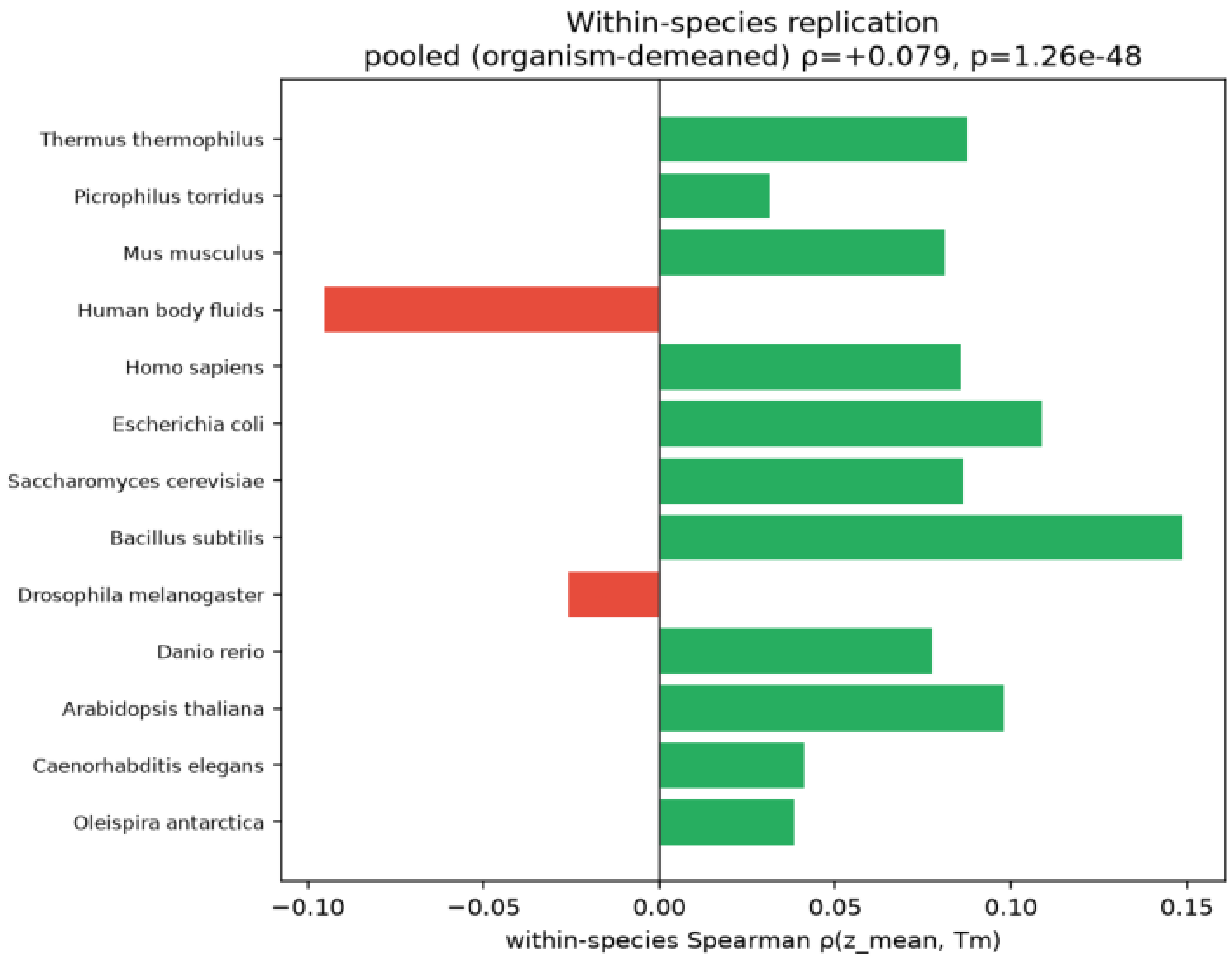
▶ H1 NOT supported: no significant cross-species OGT trend



LOO ρ collapses to zero without thermophiles

- Drop Picrophilus (60°C) → ρ = 0.000
- Drop Thermus (70°C) → ρ = 0.000
- All 11 mesophile-only subsets: ρ = +0.175 to +0.329
- Trend is a two-point extrapolation, not a broad signal
- Drop E. coli alone reduces ρ by 62% (anomalously low Z)

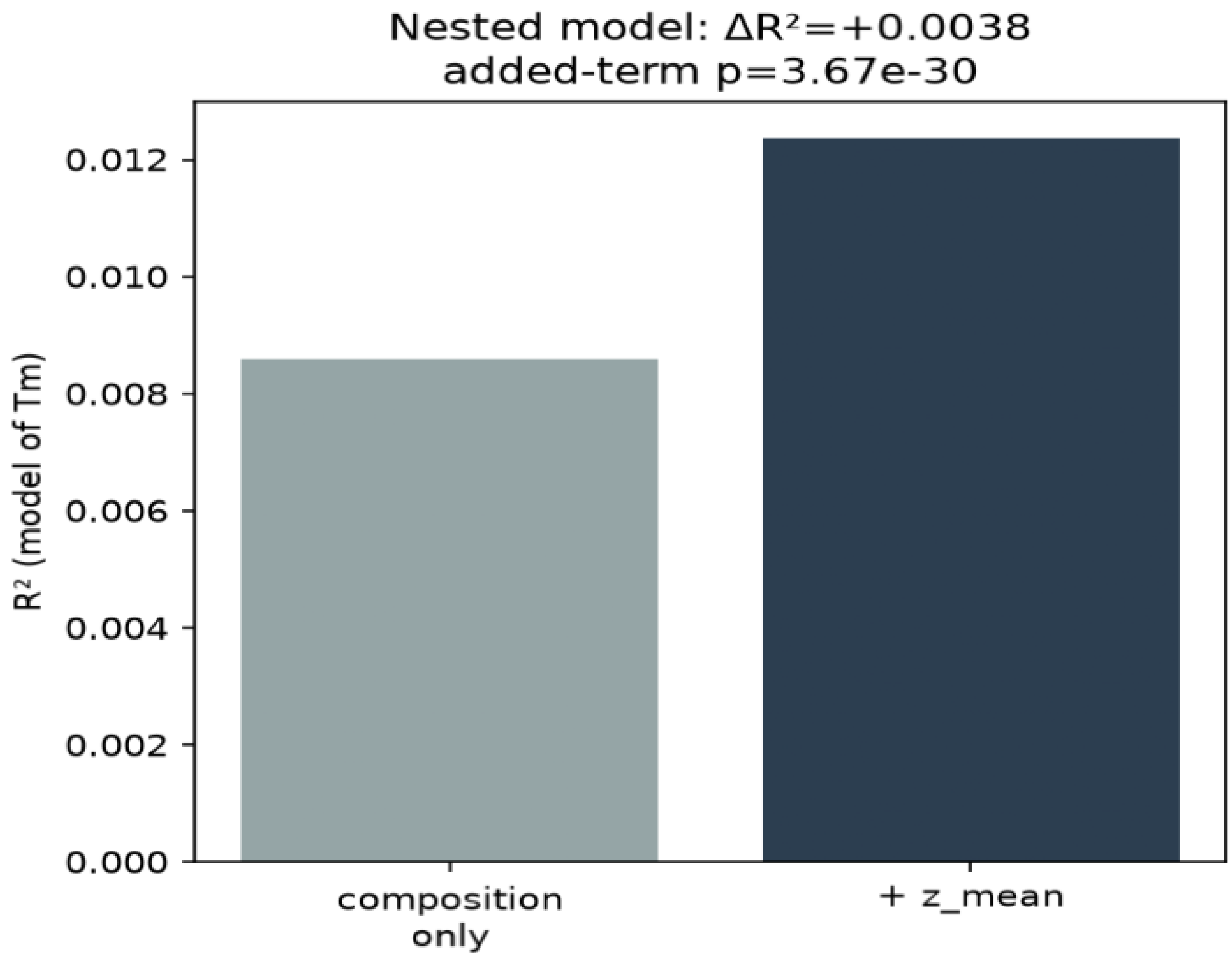
▶ **Cross-species trend depends entirely on 2 thermophiles**



Pooled $\rho = +0.079$, $p = 1.26\times10^{-48}$ (N = 34,231)

- 11 of 13 organisms show positive trend
- Bacteria strongest: B. subtilis +0.149, E. coli +0.109
- Vertebrates moderate: H. sapiens +0.086, M. musculus +0.081
- Drosophila: −0.026 (n.s.) — only negative outlier
- Psychrophile Oleispira & acidophile Picrophilus: n.s.

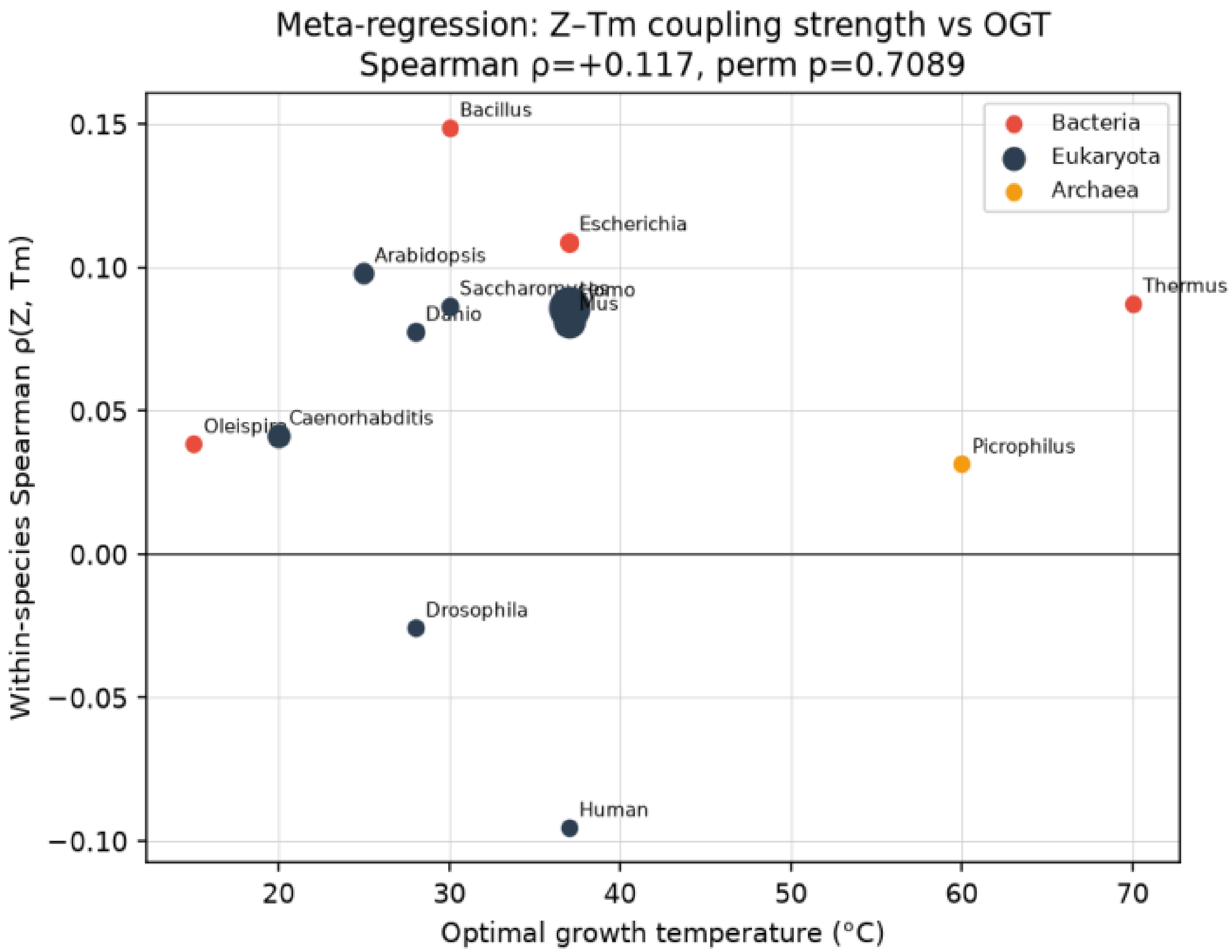
▶ **H2 SUPPORTED: within-proteome Z predicts Tm**



Z adds beyond bulk charge features

- Partial ρ(Z, OGT | composition) = +0.006, p = 0.30
- → Z carries no independent OGT signal
- Nested Tm model: $\Delta R^2 = +0.0038$, $p = 3.67\times10^{-30}$
- Semi-standardised coef: +0.688 °C / σ_Z
- Effect size small (0.38% variance) — robust at N = 34k

▶ **H3 SUPPORTED: Z–Tm is composition-independent**



Z–Tm coupling does not scale with OGT

- Meta-regression ρ(ρ_within, OGT) = +0.117, perm-p = 0.71
- Domain-level gradient (Z as stability predictor):
- Bacteria : mean ρ = +0.096 (n = 4 organisms)
- Eukaryota : mean ρ = +0.044 (n = 8 organisms)
- Archaea : mean ρ = +0.031 (n = 1 organism)

▶ **Z is most predictive for bacterial protein targets**

H1 Cross-species OGT trend — NOT SUPPORTED

- Organism-level $\rho(Z, OGT) = +0.21$, permutation $p = 0.50$ ($N = 13$ organisms).
- Signal collapses to $\rho = 0$ when either thermophile is removed (LOO test).
- After composition control: partial $\rho = +0.006$, $p = 0.30$.

H2 Within-organism Tm association — SUPPORTED

- Pooled $\rho(Z, Tm) = +0.079$, $p = 1.26\times10^{-48}$ ($N = 34{,}231$ proteins).
- Strongest in bacteria: B. subtilis +0.149, E. coli +0.109.
- Absent in Picrophilus (+0.031, n.s.) and Drosophila (−0.026, n.s.).

H3 Composition independence — SUPPORTED

- Z adds $\Delta R^2 = +0.0038$ ($p = 3.67\times10^{-30}$) beyond bulk charge features.
- Coefficient: +0.688 °C per σ_Z (semi-standardised).
- Z–Tm coupling strength does not scale with OGT (meta-regression $p = 0.71$).

Within proteomes ✓

Proteins with better-mixed surface charges (checkerboard-like Z) are measurably more thermally stable.

~0.7 °C per σ_Z, independent of total charge count. ($p = 3.67\times10^{-30}$)

Across organisms ✗

Thermophilic proteomes do NOT show higher Z than mesophiles once bulk charge density is accounted for.

OGT adaptation uses more charges, not better charge mixing.

Engineering implication: improving surface charge mixing (Z) is a real, composition-independent lever for single-protein stabilisation — particularly for bacterial enzyme targets. Combine with core-packing and disulfide strategies.